

Utilizing CdSe/ZnS Core/Shell QDs to Improve the Modulation Bandwidth of WLED for Visible Light Communication

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Abstract—Visible light communication (VLC) is an advanced and high-efficiency wireless communication technology. As one of the most important light sources in VLC, conventional white LED based on $\text{Y}_3\text{Al}_5\text{O}_{12}:\text{Ce}^{3+}$ (YAG:Ce) phosphor limits the system transmitting rate severely due to its narrow modulation bandwidth. Considering the short fluorescent lifetime of quantum dots (QDs), QD-LEDs with wide modulation bandwidth are designed here to improve the transmitting rate of VLC. CdSe/ZnS core/shell QDs and related luminescent microsphere (LMS) are implemented as light-conversion material for the QD-LEDs. Compared to conventional white LED, the proposed QD-LED reached maximum improvement on modulation bandwidth at 74.19%. Our findings will provide an effective solution of white LEDs for high speed VLC.

Keywords—quantum dots; modulation bandwidth; WLED; visible light communication;

I. INTRODUCTION

Visible light communication (VLC) is an advanced wireless communication technology that can achieve LED lighting and optical communications together by modulating LED light source [1]. Considering its advantages including high transmitting rate, low energy consumption and enhanced security, this technology has garnered a lot of attention these years in data communications [2-3]. However, as one of the most important light sources in VLC, conventional white LED (WLED) based on $\text{Y}_3\text{Al}_5\text{O}_{12}:\text{Ce}^{3+}$ (YAG:Ce) phosphor has long fluorescent lifetime, which results in a narrow modulation bandwidth and, as a consequence, the system transmitting rate has been greatly limited [4]. Therefore, how to increase the bandwidth of the light source has become a key issue.

In recent years, many studies have been performed on this problem to make further improvement of the transmitting rate for VLC system. Normally, there are mainly three methods reported in previous literatures, which could be summarized as following: Optimizing the structure of LED chip, using blue-filtering and utilizing equalization circuit as well. Optimizations on the LED chip structure aim at reducing the Resistance-Capacitance time constant and recombination time of carriers, and consequently, improving the bandwidth of

blue LED chip [5-7]. Blue-filtering is always applied to indoor VLC with WLED, which could efficiently remove the undesired signals produced by yellow phosphor [8-9]. Equalization circuit could reduce the signal distortions by resistor, inductance and resonant capacitor [10-12]. According to these researches, optimizations on structure of LED and external accessories are mainly considered but light conversion material is always ignored. As one of the most important parts in WLED, traditional light conversion material (e.g. commercial YAG:Ce phosphor) narrows down the bandwidth prominently and this problem need to be solved urgently.

Nowadays, quantum dots (QDs) have emerged as a promising material in lighting as well as display technology due to their tunable emission wavelength, narrow full width at half maximum (FWHM), nearly perfect photoluminescence (PL) and high quantum yield [13-16]. In particular, as nanocrystals, QDs could well confine the excitons in narrow distributed recombination channels, as a result of that, QDs have short fluorescent lifetime, which is far less than that of commercial YAG:Ce phosphor [17]. For light source, therefore, QD-LED could improve not only the color quality but also the bandwidth of WLED to obtain a better lighting and communication performance in VLC.

In this work, we have obtained CdSe/ZnS core/shell QD composites, luminescent micro-sphere (LMS), with different emission peak including 525nm as well as 625 nm, to manufacture green QD-LED, red QD-LED and more importantly, white QD-LED (WLED), which can be greatly practical in VLC, and then investigated the modulation bandwidths of them. According to the modulation bandwidth measuring results, QD-LED shows the better performance on VLC. Compared to traditional YAG:Ce phosphor based WLED, the enhancement for modulation bandwidth is 74% by using Red QD -LED, which is the highest improvement, meanwhile, green QD-LED and white QD-LED also have a good performance on modulation bandwidth with 64.51% and 67.74% improvement, which are far better than YAG:Ce phosphor based WLED. Combining with fluorescent lifetime results, it can be found that the red QDs, which have a shorter lifetime than green QDs, leads to a higher modulation

bandwidth improvement. Therefore modulation bandwidth of QD-LED can be improved further by utilizing QDs with shorter fluorescent lifetime.

II. EXPERIMENT

A. Materials

In this work, we obtained red and green CdSe/ZnS core/shell quantum dots. Fig. 1 shows the absorption spectra and the PL emission spectra of red QDs and green QDs. It can be found that the emission peak of red QDs and green QDs is 623 nm and 523 nm, which corresponds to red light and green light respectively. On the other hand, there is no distinct absorption peak in this figure. In this work, we obtained red and green CdSe/ZnS core/shell QDs. Meanwhile, as can be seen in Fig. 1, the FWHM is 36 nm and 40 nm for red and green QDs respectively. In addition, the QY of these two types of QDs reached to 70%.

To improve the operational performance of QDs and maintain large extent luminescent properties of QDs in WLED, QDs luminescent microspheres (LMS) have been fabricated. To obtain it, red-emitting QDs are incorporated in mesoporous silica microspheres (MS) by a non-chemical process. Owing to the lattice structure of LMS, QDs were effectively protected from intrinsic aggregation in matrix and surface deterioration by encapsulant, oxygen and moisture. The LMS composite has therefore maintained large extent luminescent properties of QDs, especially for the high quantum efficiency.

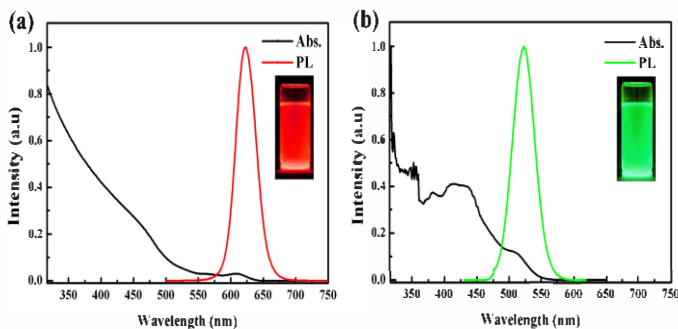


Fig. 1. Spectra of red QDs (left) and green QDs (right).

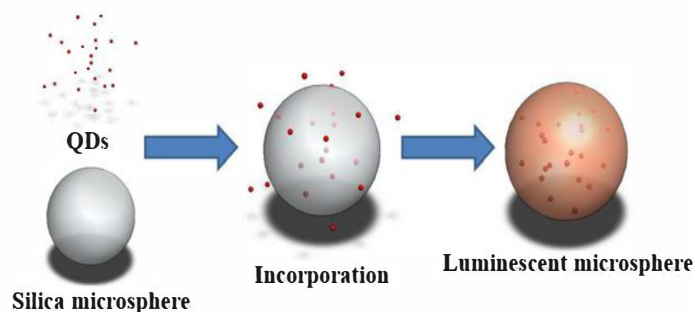


Fig. 2. Principles of obtaining luminescent microsphere.

B. LEDs fabrication and bandwidth measuring process

LEDs based on YAG:Ce phosphor, LMS and red or green QDs have been designed in our work. First of all, a blue LED packaging module, which consists of leadframe, LED chip and related components, was mounted on the substrate. Then, silicone layers mixed with LMS or YAG:Ce phosphor are fabricated. In addition, a thin-walled glass container with QDs solution in it was used to achieve QDs layers. As is shown in Fig. 3, to fabricate LEDs, layers based on phosphor, LMS and QDs were mounted on the blue LED packaging module by silicone. In LED bandwidth measuring, this structure ensures the possibility of using different layers with the same blue LED packaging module. Therefore, the instrumental error could be minimized.

A bandwidth measuring system has been set up, which includes a direct current (DC) stabilized power supply, a signal generator, a photodetector (THORLABS PDA36A) and an oscilloscope. To reduce the instrumental error, firstly DC power was applied on each LED so optical signals sent by these LEDs would be received by photodetector. Adjusting the working voltage for every LED until photodetector generated the same current. Then, in bandwidth measuring, signal generator produced a sine wave output with those above-mentioned voltages as DC bias. While signal generator was producing signals and LED was sending them, photoelectric detector could receive and show them in oscilloscope. Initially the frequency of sine wave was set at 100 KHz and the peak-to-peak voltage was set at 500mV. Then the peak-to-peak value of the sine wave was decreased as we increased the frequency of the sine wave. Once the peak-to-peak value dropped to 0.707 times of its initial value, the corresponding frequency was considered as the 3dB bandwidth of LED.

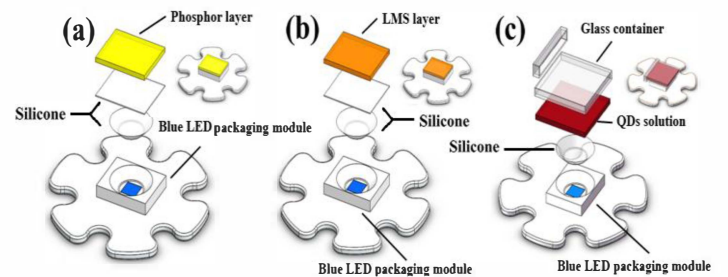


Fig.3. The structure schematic of LEDs based on (a) YAG:Ce phosphor; (b) LMS; (c) QDs

III. RESULTS AND DISCUSSIONS

A. Analyses of LED based on quantum dots solution

Fig. 4 shows the LED based on red QDs solution, green QDs solution and mixture of them. As is shown in Fig. 4(a) and 4(b), red and green light can be observed. Meanwhile,

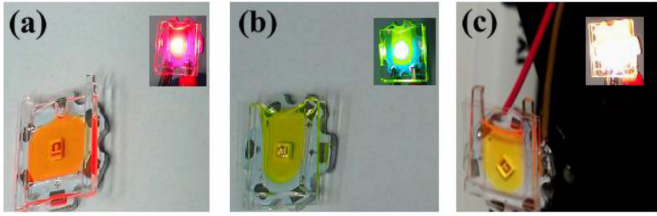


Fig. 4. LED structures with: (a) red QDs solution; (b) green QDs solution (c) mixture solution of green and red QDs;

QD-WLED can be obtained by controlling the mixing ratio of red and green QDs, which is showed in Fig. 5(c).

Fig. 5 shows the frequency responses and the bandwidth results of LEDs based on quantum dots solution. Comparing to ordinary YAG:Ce phosphor, the improvement for bandwidth is 74.19% by using red QDs, which is the highest improvement in these samples. Meanwhile, green QD-LED and mixture QD-WLED also have a nice performance on bandwidth for 64.52% and 67.75% improvements, which are far better than phosphor WLED. On the other hand, the GaN LED chip has the best performance on frequency responses and the 3dB modulation bandwidth of it reaches to 3.65MHz. GaN LED chip has the optimum performance since there is no fluorescent material on this chip. Theoretically, the modulation bandwidth of it is only determined by the minority carrier lifetime, τ , in the light-emitting 'active' region. We can obtain the theoretical bandwidth of GaN LED chips by the following function:

$$f_{3dB} = \frac{1}{2\pi\tau} \quad (1)$$

Considering the typical τ , which is greater than 100 ps in III-V materials, the bandwidth of GaN LED chips are always less than 2 GHz. Ideally, the highest bandwidth of GaN LED chips could only reached to 2 GHz in the limiting case. In fact, by frequency response measuring, the bandwidth of GaN chips are always less than 10MHz due to the disturbance caused by circuit elements, but this value is still very high respecting to LEDs with fluorescent materials.

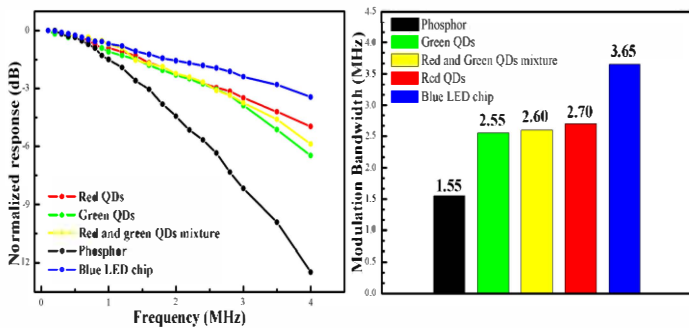


Fig. 5. Frequency responses (left) and bandwidth results(right) of measured LEDs based on QDs solution.

B. Analysis of LED based on phosphor and LMS.

Fig. 6 shows LED based on YAG:Ce phosphor with yellow light as well as LMS layer with amaranth light and Fig. 7

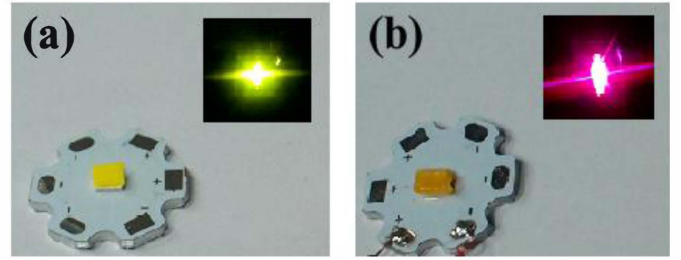


Fig. 6. LED structures with: (a) YAG phosphor layer; (b) red LMS layer.

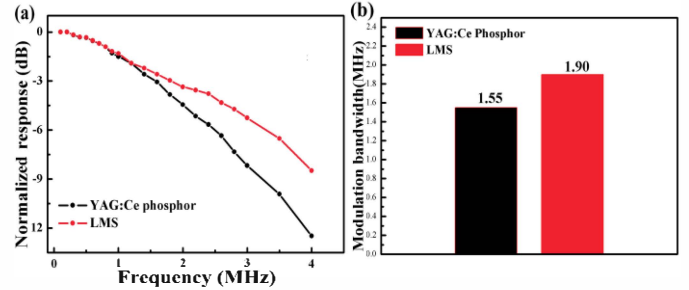


Fig. 7. (a) Normalized frequency responses of LEDs based on LMS and phosphor; (b) bandwidth results .

shows the frequency responses and the bandwidth results of LEDs based on LMS and phosphor. In this structure, YAG:Ce phosphor and LMS were mixed with silicone to fabricate layers. By this way, the light could be adjusted to be darker or lighter by controlling the material concentration and the layer thickness.

As is shows in Fig. 7, comparing with YAG:Ce phosphor, utilizing LMS as light conversion material for LEDs can increase the bandwidth by 22.58%, this value is not as good as that of QDs solution. However, since LMS are solid powers and also have the operational performance of QDs, it provide a more practical and stable way to achieve QD-LED than QDs solution.

C. Analyses on the relationship between fluorescent lifetime and bandwidth.

Fig. 8 shows that the fluorescent lifetime of red QDs (26.31 ns) is shorter than that of green QDs (29.34 ns). Because of the smearing, light conversion material limit the bandwidth of LED. On the other hand, fluorescent lifetime of light conversion material is tightly linked to the generating of smearing. To establish the relationship between fluorescent lifetime and bandwidth, modeling of the smearing has been analyzed here.

Fig. 9 shows the principles of smearing. The ideal rectangular wave transfers by visible light communication system is shown in Fig. 9. This rectangular wave signal can be observed in oscilloscope due to the photodetector receives this LED optical signal and transforms it into current signal, then, the oscilloscope converts it into volatge signal and displays it. For phosphor WLED, due to the limitation of light-conversion material's fluorescent lifetime, it requires some time for signal from high voltage to low voltage so smearing is generated as red lines show in Fig. 9. As Fig. 9(c) and Fig. 9(d) show, it can be easily found that for the same fluorescent lifetime, the

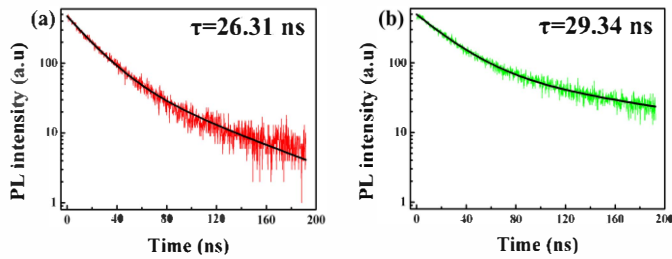


Fig. 8. The fluorescent lifetime of (a) red QDs; (b) green QDs.

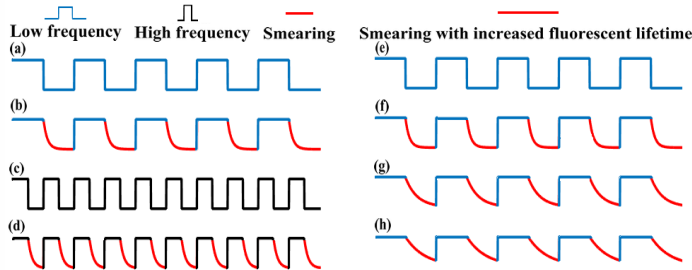


Fig. 9. The principles of smearing: (a) low frequency signal; (b) low frequency signal with smearing; (c) high frequency signal; (d) high frequency signal with smearing; (e)-(h) signal with smearing that more and more severe.

the frequency of signal is, the badly the smearing is, thus, the signal distortion is more serious and the bandwidth is more narrow. On the other hand, as fig. 9(e)-9(h) show, for the same signal frequency, the longer the fluorescent lifetime is, the longer the time from high voltage to low voltage needs, therefore, the signal distortion is more serious and the bandwidth is more narrow.

According to these analyses, we can infer that, QD-LED has a higher bandwidth than LED based on YAG:Ce since phosphor has a far longer fluorescent lifetime. Furthermore, LED based on red QDs, which has a shorter fluorescent lifetime, provides a higher bandwidth than the LED based on green QDs, meanwhile, the white QD-LED, which consists of red and green QDs, will has a moderate bandwidth between that of those two types of QDs. Obviously, these predictions can be verified by Fig. 5. As for LMS, since it is formed by QDs and LMS, it will also have a higher bandwidth than traditional WLED, which can be verified by Fig. 7. In addition, it can be predicted that the higher the signal frequency is, the better the performance of QD-LED is. This characteristic can also be confirmed by Fig. 5 and Fig. 7, which indicates the noise of QD-LED is far lower than phosphor LED when the frequency is high.

IV. SUMMARY

QD-LEDs that based on green QDs, red QDs and related LMS were successfully fabricated, then the modulation bandwidth measurement for them were implemented. Comparing to the traditional LED that based on YAG:Ce phosphor, the proposed QD-WLED and QD-LED reached maximum improvement on modulation bandwidth at 67.75% and 74.19% respectively. This result indicated the potential advantages of utilizing QDs LED in VLC are various, including better luminous effect, lower energy consumption

and higher transmission rate due to the wider bandwidth. Furthermore, the mechanism that how the fluorescent lifetime affects the bandwidth of QD-LEDs was also discussed, due to the smearing effect, a shorter fluorescent lifetime leads to a higher modulation bandwidth. Our findings can help building up the visible light communication system with better performance in lighting and signals transmission.

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